
Introductory Statistics for Econometrics

Unit 6. Special Distributions, including the Multivariate Normal

Bernoulli, Binomial, Poisson, Uniform, Exponential,
Normal, χ^2 , t , F , and the multivariate normal

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Learning outcomes. After this unit, the student should be able to (i) recognise and use the main parametric families of distributions; (ii) standardise a random variable and use the standard normal tables; (iii) state the DeMoivre–Laplace theorem as a preview of the CLT; (iv) build the χ^2 , t , and F distributions from i.i.d. normals; (v) work with the multivariate normal $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$.

Source material: condensed from MIT OCW 14.30 (Spring 2009), K. Menzel, Lecture Notes 14, augmented with the multivariate normal.

1 Standardisation

For any random variable X with $\mu = \mathbb{E}[X]$ and $\sigma^2 = \text{Var}(X) > 0$, the *standardisation*

$$Z = \frac{X - \mu}{\sigma}$$

satisfies $\mathbb{E}[Z] = 0$ and $\text{Var}(Z) = 1$. Standardisation removes location and scale, allowing us to compare shapes of distributions and to apply universal tables (such as the standard normal table).

2 Discrete benchmarks

Definition 2.1 (Bernoulli). $X \sim \text{Bernoulli}(p)$ if $X \in \{0, 1\}$ with $\mathbb{P}(X = 1) = p$. $\mathbb{E}[X] = p$, $\text{Var}(X) = p(1 - p)$.

Definition 2.2 (Binomial). $X \sim B(n, p)$ if it is the number of successes in n i.i.d. Bernoulli(p) trials: $f_X(x) = \binom{n}{x} p^x (1 - p)^{n-x}$, $x \in \{0, \dots, n\}$. $\mathbb{E}[X] = np$, $\text{Var}(X) = np(1 - p)$.

Definition 2.3 (Poisson). $X \sim \text{Poisson}(\lambda)$ if $f_X(x) = e^{-\lambda} \lambda^x / x!$, $x = 0, 1, 2, \dots$. $\mathbb{E}[X] = \text{Var}(X) = \lambda$.

When to use Poisson. Counts of rare events with constant intensity: arrivals at a queue, defaults in a portfolio, accidents per month, citation counts. Sums of independent Poissons are Poisson: $\text{Poisson}(\lambda_1) + \text{Poisson}(\lambda_2) = \text{Poisson}(\lambda_1 + \lambda_2)$.

Binomial-to-Poisson approximation. If $n \rightarrow \infty$, $p \rightarrow 0$ with $np \rightarrow \lambda$ fixed, then $B(n, p) \rightarrow \text{Poisson}(\lambda)$. This is why Poisson is the natural model for “many trials, small probability of success in each”.

3 Continuous benchmarks

Definition 3.1 (Uniform). $X \sim U[a, b]$ if $f_X(x) = 1/(b - a)$ on $[a, b]$. $\mathbb{E}[X] = (a + b)/2$, $\text{Var}(X) = (b - a)^2/12$.

Definition 3.2 (Exponential). $X \sim \text{Exp}(\lambda)$ if $f_X(x) = \lambda e^{-\lambda x}$ on $x \geq 0$. $\mathbb{E}[X] = 1/\lambda$, $\text{Var}(X) = 1/\lambda^2$.

The exponential is the unique continuous distribution that is *memoryless*: $\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t)$. It is the natural model for waiting times under constant hazard, and it is the inter-arrival distribution of a Poisson process.

4 The normal/Gaussian distribution

Definition 4.1 (Standard normal). $Z \sim \mathcal{N}(0, 1)$ has p.d.f.

$$\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad z \in \mathbb{R},$$

and c.d.f. $\Phi(z) = \int_{-\infty}^z \varphi(s) ds$ (no closed form; tabulated). Properties: $\mathbb{E}[Z] = 0$, $\text{Var}(Z) = 1$, $\varphi(-z) = \varphi(z)$, $\Phi(-z) = 1 - \Phi(z)$.

Definition 4.2 (General normal). $X \sim \mathcal{N}(\mu, \sigma^2)$ if $X = \mu + \sigma Z$ for some $Z \sim \mathcal{N}(0, 1)$. Its p.d.f. is

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right].$$

Two stability properties.

(1) Affine: $X \sim \mathcal{N}(\mu, \sigma^2) \Rightarrow a + bX \sim \mathcal{N}(a + b\mu, b^2\sigma^2)$.

(2) Independent sums: $X_i \overset{\perp}{\sim} \mathcal{N}(\mu_i, \sigma_i^2) \Rightarrow \sum_i X_i \sim \mathcal{N}(\sum_i \mu_i, \sum_i \sigma_i^2)$.

The normal family is therefore closed under both linear transformations and independent addition — a unique combination among standard distributions.

The 68–95–99.7 rule.

$$\mathbb{P}(|X - \mu| \leq \sigma) \approx 68\%, \quad \mathbb{P}(|X - \mu| \leq 2\sigma) \approx 95\%, \quad \mathbb{P}(|X - \mu| \leq 3\sigma) \approx 99.7\%.$$

Useful as a sanity check whenever you read a confidence interval or a z -statistic.

Theorem 4.3 (DeMoivre–Laplace). If $X_n \sim B(n, p)$ then for every $a < b$,

$$\lim_{n \rightarrow \infty} \mathbb{P}\left(a \leq \frac{X_n - np}{\sqrt{np(1-p)}} \leq b\right) = \Phi(b) - \Phi(a).$$

The DeMoivre–Laplace theorem is the historical ancestor of the central limit theorem (Unit 8). Numerically, it gives normal approximations to binomial probabilities that are extremely accurate for moderate n when p is not too close to 0 or 1.

Computation with Φ . For $X \sim \mathcal{N}(\mu, \sigma^2)$ and $a < b$,

$$\mathbb{P}(a \leq X \leq b) = \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right).$$

Tables typically only list Φ for non-negative arguments; use $\Phi(-z) = 1 - \Phi(z)$ for the rest.

Example 4.4. $\tilde{X} \sim B(100, 0.2)$. Standardising, $Z = (\tilde{X} - 20)/4$.

$$\mathbb{P}(\tilde{X} \leq 25) \approx \Phi(5/4) = \Phi(1.25) \approx 0.894.$$

The exact binomial probability is 0.913, so the approximation has an error of about 2 percentage points at $n = 100$, and of less than 0.5 points at $n = 200$.

5 Distributions built from normals: χ^2 , t , F

These three families are not interesting in themselves but as *sampling distributions* of statistics we will compute in later units.

Definition 5.1 (Chi-squared). If $Z_1, \dots, Z_k \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$, then $Y = \sum_{i=1}^k Z_i^2 \sim \chi_k^2$ (chi-squared with k degrees of freedom). $\mathbb{E}[Y] = k$, $\text{Var}(Y) = 2k$.

Definition 5.2 (Student's t). If $Z \sim \mathcal{N}(0, 1)$ and $Y \sim \chi_k^2$ are independent, then

$$T = \frac{Z}{\sqrt{Y/k}} \sim t_k.$$

$\mathbb{E}[T] = 0$ if $k > 1$ not defined for other values; $\text{Var}(T) = k/(k-2)$ if $k > 2$ not defined for other values. For large k , $t_k \rightarrow \mathcal{N}(0, 1)$.

Definition 5.3 (F-distribution). If $Y_1 \sim \chi_{k_1}^2$ and $Y_2 \sim \chi_{k_2}^2$ are independent,

$$F = \frac{Y_1/k_1}{Y_2/k_2} \sim F(k_1, k_2).$$

Where these will appear.

- χ_{n-1}^2 is the sampling distribution of $(n-1)S^2/\sigma^2$ when X_i are i.i.d. normal (Unit 9).
- t_{n-1} is the sampling distribution of $(\bar{X} - \mu)/(S/\sqrt{n})$ — the basis of the t -test (Unit 10).
- F is the sampling distribution of ratios of variance estimators and the basis of the joint significance test in OLS regression (also Unit 10).

6 The multivariate normal

In Unit 6 we introduced random vectors with mean $\boldsymbol{\mu}$ and covariance matrix Σ . The multivariate normal is the natural generalisation of the univariate normal to such vectors.

Definition 6.1 (Multivariate normal). A random vector $\mathbf{X} \in \mathbb{R}^n$ has the *multivariate normal distribution* $\mathcal{N}_n(\boldsymbol{\mu}, \Sigma)$ with $\boldsymbol{\mu} \in \mathbb{R}^n$ and $\Sigma \succ 0$ if its joint density is

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left[-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right].$$

For $n = 2$, with $\boldsymbol{\mu} = (\mu_1, \mu_2)^\top$, $\sigma_i^2 = \text{Var}(X_i)$, $\rho = \text{Corr}(X_1, X_2)$, the level sets are *ellipses* centred at $\boldsymbol{\mu}$, with axes determined by the eigenvectors of Σ .

Proposition 6.2 (Five essential properties). Let $\mathbf{X} \sim \mathcal{N}_n(\boldsymbol{\mu}, \Sigma)$.

- (1) Affine transforms are normal. For any constants $A \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$, $A\mathbf{X} + \mathbf{b} \sim \mathcal{N}_m(A\boldsymbol{\mu} + \mathbf{b}, A\Sigma A^\top)$.
- (2) Marginals are normal. Each $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$; more generally any subvector of $\mathbf{X}$ is multivariate normal.
- (3) Conditionals are normal. Partition $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2)^\top$ with means $(\boldsymbol{\mu}_1, \boldsymbol{\mu}_2)$ and covariance $\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$. Then

$$\mathbf{X}_1 \mid \mathbf{X}_2 = \mathbf{x}_2 \sim \mathcal{N}(\boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}).$$

- (4) Independence = zero correlation. Within a multivariate normal, $\text{Cov}(X_i, X_j) = 0$ if and only if $X_i \perp X_j$. (False for non-Gaussian vectors — recall Unit 6.)
- (5) Standardisation. If $\Sigma \succ 0$ and $\mathbf{Z} = \Sigma^{-1/2}(\mathbf{X} - \boldsymbol{\mu})$, then $\mathbf{Z} \sim \mathcal{N}_n(\mathbf{0}, I_n)$.

The CEF in the bivariate case. For $(X, Y) \sim \mathcal{N}_2$ with means μ_X, μ_Y , variances σ_X^2, σ_Y^2 and correlation ρ , property (3) gives

$$\mathbb{E}[Y \mid X = x] = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X), \quad \text{Var}(Y \mid X = x) = (1 - \rho^2) \sigma_Y^2.$$

The conditional expectation function is exactly linear in x , with slope $\text{Cov}(X, Y)/\text{Var}(X)$ — the same population OLS coefficient we derived in Unit 6, now *justified* by the assumption of joint normality.

Why the multivariate normal is everywhere.

- Sums of i.i.d. random vectors converge to a multivariate normal (Multivariate CLT, Unit 8).
- OLS coefficient estimators are asymptotically multivariate normal under standard regularity conditions.
- Joint hypothesis tests on regression coefficients (the F -test of Unit 10) reduce to chi-squared distances under multivariate normal sampling.

Summary

- Discrete: Bernoulli(p), Binomial(n, p), Poisson(λ). Sums of independent same-family variables are in the family for Binomial (same p) and Poisson.
- Continuous benchmarks: $U[a, b]$, $\text{Exp}(\lambda)$ (memoryless).
- $\mathcal{N}(\mu, \sigma^2)$: closed under affine transforms and independent addition; 68/95/99.7 rule for sanity checks; DeMoivre–Laplace previews the CLT.
- Sampling distributions built from normals: $\chi_k^2 = \sum Z_i^2$, $t_k = Z/\sqrt{Y/k}$, $F(k_1, k_2) = (Y_1/k_1)/(Y_2/k_2)$.
- Multivariate normal $\mathcal{N}_n(\boldsymbol{\mu}, \Sigma)$: affine, marginals and conditionals are all multivariate normal; zero correlation $\Leftrightarrow$ independence; quadratic form $\sim \chi_n^2$ after standardisation.

Suggested exercises

1. Use the standard normal table to compute (a) $\mathbb{P}(Z > 1.96)$, (b) $\mathbb{P}(-1.64 \leq Z \leq 1.64)$, (c) $\mathbb{P}(0.5 \leq Z \leq 2)$.
2. Let $X \sim \mathcal{N}(100, 225)$. Compute $\mathbb{P}(X \leq 110)$, $\mathbb{P}(85 \leq X \leq 120)$, and the 95th percentile of X .
3. Sum of normals. $X \sim \mathcal{N}(2, 4)$, $Y \sim \mathcal{N}(-1, 9)$ independent. Find the distribution of $X - Y$ and of $(X + Y)/2$.
4. Show that if $Z \sim \mathcal{N}(0, 1)$ then $Z^2 \sim \chi_1^2$ by a direct change-of-variables calculation.
5. Bivariate normal CEF. $(X, Y) \sim \mathcal{N}_2$ with $\mu_X = \mu_Y = 0$, $\sigma_X^2 = \sigma_Y^2 = 1$, $\rho = 0.6$. Compute $\mathbb{E}[Y \mid X = 1.2]$ and the 95% predictive interval for Y given $X = 1.2$.
6. Independence vs. zero correlation. Construct two random variables X, Y that are uncorrelated but not independent (recall Unit 6). Then show that within a bivariate normal this is impossible.